# Multimedia Appendix 1: Search strategy and data extraction form

## Databases:

Pubmed, IEEE Xplore, EMBASE, Web of Science, ACM CHI, Cochrane Library

## Date:

January 2007 to September 2021

## Language:

English

## Search strings:

**PubMed (https://pubmed.ncbi.nlm.nih.gov/)**

(eHealth OR mHealth OR "digital health" OR "remote patient monitoring" OR "remote monitoring technology" OR "remote measurement technology" OR telehealth OR telemonitoring OR application OR wearable) AND (patient AND (data OR symptom*) AND (visual OR visualiz* OR visualis* OR present* OR graph*) AND (prefer* OR perception OR usability OR design OR acceptability)) AND ("central nervous system" OR "nervous system" OR psychiat* OR neurolog* OR neurodegen* OR "mental health")

**IEEE Xplore (https://ieeexplore.ieee.org/Xplore/home.jsp)**

(((((("Abstract":ehealth OR mhealth OR "digital health'" OR "remote patient monitoring" OR "remote monitoring technology" OR "remote measurement technology" OR telehealth OR telemonitoring OR application OR wearable) AND "Abstract":visual OR visualiz* OR visualis* OR present* OR graph*) AND "Abstract":"central nervous system" OR "nervous system" OR psychiatric OR psychiatry OR neurology OR neurological OR neurodegeneration OR "mental health") AND "Abstract":Preference OR preferences OR preferred OR perception OR usability OR design OR acceptability) AND "Abstract":patient AND (data OR symptom OR symptoms))

**EMBASE* (https://www.embase.com/)**

(ehealth OR mhealth OR 'digital health' OR 'remote patient monitoring' OR 'remote monitoring technology' OR 'remote measurement technology' OR telehealth OR telemonitoring OR application OR wearable) AND patient AND (data OR symptom*) AND (visual OR visualiz* OR visualis* OR present* OR graph*) AND (prefer* OR perception OR usability OR design OR acceptability) AND ('central nervous system' OR 'nervous system' OR psychiat* OR neurolog* OR neurodegen* OR 'mental health') AND [2007-2020]/py AND [humans]/lim AND [english]/lim AND 'article'/it

**Web of Science* (https://login.webofknowledge.com/)**

(ALL = ((ehealth OR mhealth OR "digital health'" OR "remote patient monitoring" OR "remote monitoring technology" OR "remote measurement technology" OR telehealth OR telemonitoring OR application OR wearable) AND patient* AND (data OR symptom*) AND (visual OR visualiz* OR visualis* OR present* OR graph*) AND (prefer* OR perception OR usability OR design OR acceptability) AND ("central nervous system" OR "nervous system'"OR psychiat* OR neurolog* OR neurodegen* OR "mental health"))) AND LANGUAGE: (English) AND DOCUMENT TYPES: (Article) Indexes=SCI-EXPANDED, SSCI, A&HCI, ESCI Timespan=2007-2020

**ACM* (https://dl.acm.org/)**

(eHealth OR mHealth OR "digital health" OR "remote patient monitoring" OR "remote monitoring technology" OR "remote measurement technology" OR telehealth OR telemonitoring OR application OR wearable) AND (patient AND (data OR symptom*) AND (visual OR visualiz* OR visualis* OR present* OR graph*) AND (prefer* OR perception OR usability OR design OR acceptability)) AND ("central nervous system" OR "nervous system" OR psychiat* OR neurolog* OR neurodegen* OR "mental health")

**Cochrane Library (https://www.cochranelibrary.com/)**

(eHealth OR mHealth OR "digital health" OR "remote patient monitoring" OR "remote monitoring technology" OR "remote measurement technology" OR telehealth OR telemonitoring OR application OR wearable) AND (patient AND (data OR symptom*) AND (visual OR visualiz* OR visualis* OR present* OR graph*) AND (prefer* OR perception OR usability OR design OR acceptability)) AND ("central nervous system" OR "nervous system" OR psychiat* OR neurolog* OR neurodegen* OR "mental health")

## Data extraction form

| **RefID:** | | | |
|---|---|---|---|
| **Population:** | | | **Description/Purpose of App/Study:** |
| **Type of Study** | | **N** | |
| | Exploratory qual study (no spec. app) | | |
| | User-centric design/co-design | | |
| | App usability (lab-based) | | **Data Visualized:** |
| | Field Test | | **User Acceptance Scores:** |
| | Other | | **Study Risk of Bias/Quality Assessment:** |
| **Qualitative data:** | | | **Link to annotated PDF:** |
| **Mixed methods?:** | | | **Other notes:** |

| | |
|---|---|
| screenshots | Background about screenshots |

| **Page** | **Data Visualization: Extracted Data/Quotes for coding(context in italics)** | **Codes** |
|---|---|---|
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